

MTH202 MID TERM SOLVED MCQS DOWNLOADED FROM VURANK

Time: 60 min

Marks: 40

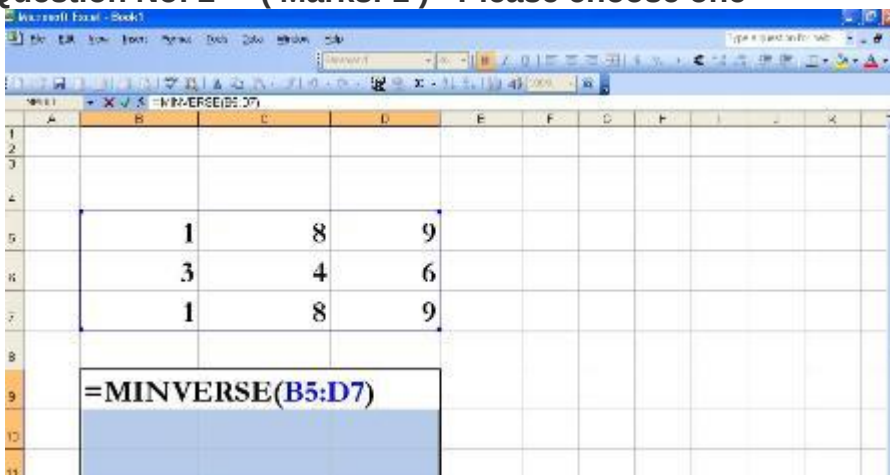
Question No: 1 (Marks: 1) - Please choose one

In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- ▶ Provident Fund
- ▶ **Gratuity Fund**
- ▶ Charity Fund
- ▶ None of the above

Gratuity Fund: According to local laws, a company can establish a Gratuity Trust Fund for the benefit of the employees. By law, 1/11th of Basic Salary per month is contributed by the company to the Gratuity Fund to the account of the employee. Thus there is a saving of 1/11th of basic salary on behalf of the employee in Gratuity Fund.

Question No: 2 (Marks: 1) - Please choose one



	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5		1	8	9							
6		3	4	6							
7		1	8	9							
8											
9		=MINVERSE(B5:D7)									
10											
11											

The answer of the above formula is

- ▶ None of the above.

▶ a number

▶ a matrix

▶ #NUM!

MINVERSE: Returns the inverse matrix for the matrix stored in an array.

Question No: 3 (Marks: 1) - Please choose one

If a profit ratio of A and B is 2 to 3, then A's share in profit will be

▶ 2/5

▶ 2/3

▶ 3/2

▶ 3/5

Question No: 4 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

▶ Sale price

▶ Revenue discount

▶ Selling price

▶ Cost price

Question No: 5 (Marks: 1) - Please choose one

A reduction of the amount due on an invoice is called a _____ .

▶ Trade discount

▶ Net discount

► Cash discount

► Unearned discount

Trade discount represents a reduction in list price in return for quantity purchases.

Question No: 6 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

► 1 / 11 th

► 2 / 11 th

► 9.09 %

► 9.99%

Provident Fund 9.09 % of basic salary.

Question No: 7 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 2 & 6 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix}$$

, then AB is -----

► $\begin{bmatrix} 20 & 36 \\ 25 & 30 \end{bmatrix}$

► $\begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$

$$\blacktriangleright \begin{bmatrix} 20 & 36 \\ 35 & 30 \end{bmatrix}$$

$$\blacktriangleright \begin{bmatrix} 20 & 36 \\ 45 & 39 \end{bmatrix}$$

Solution:

$$AB = 2 \times 1 + 6 \times 3 \quad 2 \times 9 + 6 \times 3$$

$$3 \times 1 + 4 \times 3 \quad 3 \times 9 + 4 \times 3$$

$$AB = 2 + 18 \quad 18 + 18$$

$$3 + 12 \quad 27 + 12$$

$$AB = 20 \quad 36$$

$$15 \quad 39$$

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Question No: 8 (Marks: 1) - Please choose one

$A = \begin{bmatrix} 1 \end{bmatrix}$, then which statement is wrong about A?

☐ A is a column matrix.

☐ A is a row matrix.

☐ A is an identity matrix.

☒ A is not a square matrix.

Question No: 9 (Marks: 1) - Please choose one

If a speed of a car is changed from 25km/h to 40km/h then the percentage change in its speed is?

► 15%

► 50%

► 40%

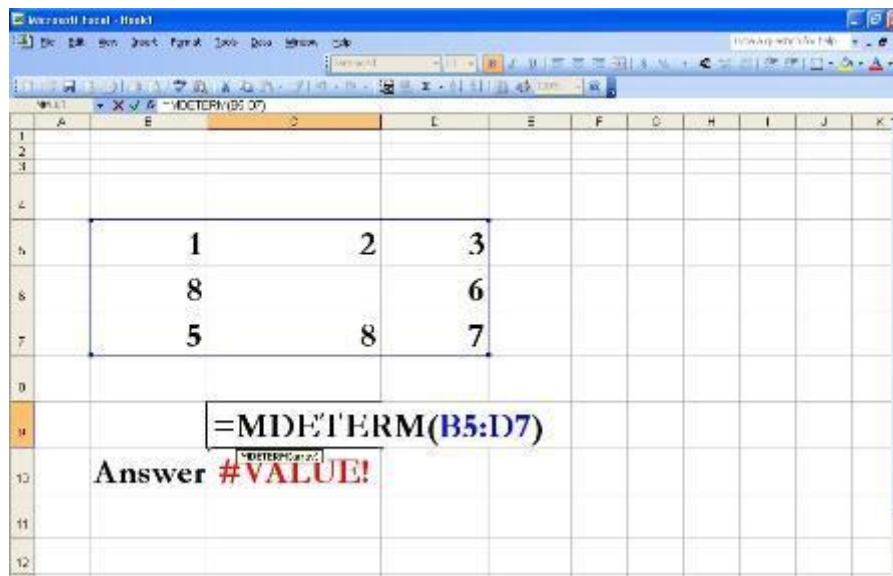
► 60%

Solution:

$$25-40=15$$

$$15/25 \times 100 = 60$$

Question No: 10 (Marks: 1) - Please choose one



The answer of the above formula is #VALUE!. The reason of this answer is

► Inverse of the matrix does not exist.

► It has one blank cell.

► Number of rows and columns are equal.

► The keys CTRL+SHIFT+ENTER were not pressed simultaneously.

Question No: 11 (Marks: 1) - Please choose one

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$8,000. The useful life of the asset is 5 years. The depreciation is being calculated for the third year, and there are 10 months in the first year.

► =DB (10000, 5, 8000, 3, 10)

► =DB (10000, 8000, 5, 10, 3)

► =DB (10000, 8000, 5, 3, 10)

► =DB (10000, 8000, 10, 3, 5)

DB: Returns the depreciation of an asset for a specified period using the fixed-declining balance method.

Syntax:

DB(cost,salvage,life,period,month)

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Question No: 12 (Marks: 1) - Please choose one

This example returns the present value of an investment that pays Rs. 100 at the end of every year for 10 years. The money paid out will earn 5.25% annually.

► =PV (5.25%/1, 10*1, 100, 0)

► =PV (5.25%/1, 10*1, 100, 1)

► =PV (5.25%/12, 10*1, 100, 0)

► =PV (5.25%/1, 10*12, 100, 1)

PV: Returns the present value of an investment

Syntax: PV(rate,nper,pmt,fv,type)

Rate interest rate per period= 5.25%

Nper total number of payment periods in an annuity

Pmt payment made each period and cannot change over the life of the annuity

Fv future value, or a cash balance you want to attain after the last payment is made

Type number 0 or 1 and indicates when payments are due.

Question No: 13 (Marks: 1) - Please choose one

To add two cells (A1 and A2) together you use the following formula

▶ **=A1 + A2**

▶ =Add(A1+A2)

▶ =together(A1:A2)

▶ A1 plus A2

Question No: 14 (Marks: 1) - Please choose one

If A is the matrix of dimension 2x3 and I is the identity matrix of dimension 2x3.
Then which of the following is true

▶ $AI = I$

▶ $AI = A$

▶ $A + I = A$

▶ **$A + I = I$**

IDENTITY MATRIX: An identity matrix is a square matrix with 1's on the main diagonal from the upper left to the lower right and 0's off the main diagonal. An identity matrix is denoted as I.

Question No: 15 (Marks: 1) - Please choose one

The point at which no profit is made and no losses are incurred is said to be

- ▶ Fixed Cost point
- ▶ Contribution Margin
- ▶ **Break even point**
- ▶ Contribution Rate

Break Even point is a point at which neither a profit nor loss is made.

Question No: 16 (Marks: 1) - Please choose one

Cost analysis provides the following information:

Fixed Costs (FC) per period = Rs. 20000

Variable Costs (VC) = Rs. 30 per unit.

Selling price per unit = S = 50 Rs . Contribution Margin will be

▶ **20 Rs**

- ▶ 80 Rs
- ▶ 16 Rs
- ▶ 26 Rs

Contribution Margin= S-VC

=50-30

=20

Question No: 17 (Marks: 1) - Please choose one

If there is a change of -30% in the price of an item, what does the negative sign show?

► **The price is decreasing.**

- The price has low rate of change.
- None of the above.
- The price is increasing.

Question No: 18 (Marks: 1) - Please choose one

Net income can be calculated by using

- Net income = Number of units sale above break even point * Price per unit
- Net income = Total number of units sold * Price per unit
- Net income = Number of units sale above break even point * contribution margin per unit

► **Net income = Total number of units sold * contribution margin per unit.**

Net Income (NI) or Profit

Net income=NI=Number of units sale above BEP in units × Contribution Margin per unit

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Question No: 19 (Marks: 1) - Please choose one

If you want to multiply a number 7 to matrix

$$B = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

result is

then the

$$\blacktriangleright \begin{bmatrix} 14 & 3 \\ 28 & 1 \end{bmatrix}$$

$$\blacktriangleright \begin{bmatrix} 2 & 21 \\ 4 & 7 \end{bmatrix}$$

$$\blacktriangleright \begin{bmatrix} 14 & 21 \\ 28 & 7 \end{bmatrix}$$

► Number cannot be multiply to matrix

Solution:

$$B = 7 \times 2 \quad 7 \times 3$$

$$7 \times 4 \quad 7 \times 1$$

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Question No: 20 (Marks: 1) - Please choose one

Given : List Price = 5500Rs , Discount = 850Rs, then Net Cost Price will be

► 6350 Rs

► 5508 Rs

► 5585 Rs

► **4650 Rs**

Net Cost Price = List price – Discount

$$= 5,500 - 850$$

$$= 4,650 \text{Rs}$$

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Question No: 21 (Marks: 2)

	A	B	C	D
1				
2				
3				
4				
5				
6			2400	Cost
7			8/19/2008	Date Purchased
8			12/31/2008	End of First period
9			300	Salvage value
10			1	Period
11			0.15	Deprication rate
12			1	Actual basis
13			=AMORDEGRC(C6,C7,C8,C9,C10,C11,C12)	
14				

What is the purpose of above given function and why we have used here with respect to give data?

Answer:

AMORDEGRC Returns the depreciation of an asset, for each accounting period by using depreciation coefficient.

The formula is AMORDEGRC (Cost, start date , end date, salvage value, nper, rate, type)

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Question No: 22 (Marks: 2)

An item originally priced at \$55 is marked up 25% . What is the sale price?

Answer:

Sale Price = Original Price × (1 + Markup on original price)

Sale Price = \$ 55 × (1 + 25/100)

$$\text{Sale Price} = \$ 55 \times (1 + 0.25)$$

$$\text{Sale Price} = \$ 55 \times 1.25$$

$$\text{Sale Price} = \$ 68.75$$

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Question No: 23 (Marks: 3)

An item is marked down 17%; the sale price is \$137.46. What was the original price?

Answer:

$$\text{Selling price} = \text{Original Price} (1 - \text{markup})$$

Let suppose Original Price is S

$$137.46 = S \times (1 - 0.17)$$

$$S = 137.46 / 0.83$$

$$\text{Then Original Selling Price} = 165.61445$$

Question No: 24 (Marks: 3)

Solve the following system of equations by substitution method.

$$2x - 3y = -2, \quad 4x + y = 24$$

Answer:

Linear equations by substitution method:

$$4x + y = 24$$

$$Y = 24 - 4x$$

By putting the value of $y = 24 - 4x$, we will get:

$$2x - 3(24 - 4x) = -2$$

$$2x - 72 + 12x = -2$$

$$14x = 72 - 2$$

$$14x = 70$$

$$X = 70/14$$

$$X = 5$$

By putting the value of $x = 5$ in the following equation we will get:

$$2x - 3y = -2$$

$$2(5) - 3y = -2$$

$$10 - 3y = -2$$

$$-3y = -2 - 10$$

$$-3y = -12$$

$$Y = 12/3$$

$$Y = 4$$

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Question No: 25 (Marks: 5)

A gold chain is sold for Rs. 6500 at a gain of 25%. Find the profit.

Answer:

Sale price = Rs 6500

Cost Price = C

Gain = 25%

List Price = Sale price – (25% of Sale price)

= 6500 – 1625

= 4875Rs

Profit = 6,500 – 4,875

Profit = 1,625Rs.

Question No: 26 (Marks: 5)

A retailer used a markup rate of 18% on cost. Find the selling price of an item that costs the retailer \$60.

Answer:

Cost = \$60

Markup Rate = 18% on Cost

Selling Price = Cost Price x (1 + Markup on Cost Price)

Selling Price = 60 x (1 + 0.18)

Selling Price = 60 x 1.18

Selling Price by retailer = \$ 70.8

SHARED BY MOST WANTED

Time: 90 min

Question No: 1 (Marks: 1)

An arrangement of data by successive time periods is called a

- ▶ Exponential Smoothing
- ▶ Time Series
- ▶ Combination
- ▶ Permutation

Question No: 2 (Marks: 1)

What is the probability of choosing a vowel from the alphabet?

- ▶ 21/26
- ▶ 5/26
- ▶ 1/21
- ▶ 2/21

Question No: 3 (Marks: 1)

What is the probability of scoring 11 when you roll two dice?

- ▶ 1/18
- ▶ 2/18
- ▶ 1/36
- ▶ 3/18

Question No: 4 (Marks: 1)

If sign of r is negative then it indicates

- ▶ Direct relationship between X & Y
- ▶ Indirect relationship between X & Y
- ▶ X & Y equal
- ▶ X & Y are square

Question No: 5 (Marks: 1)

The answer of the SUMIF function in the above diagram is:

► 763

► 663

► 613

► 513

Question No: 6 (Marks: 1)

Two dice are rolled and the numbers are added together. The probability of the total being 12 is

► True

► False

Question No: 7 (Marks: 1)

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

► Mean

► Median

► Mode

► Midrange

Question No: 8 (Marks: 1)

The experimental region is the range of the previously observed values of the dependent variable.

► False

► True

Question No: 9 (Marks: 1)

----- should be of equal size.

► Intervals

► table

► frequency

► mean

Question No: 10 (Marks: 1)

Let's assume that you are receiving 1000 Rs. every year, and you invested each payment at 5%. The amount you would have at the end of five years period is referred as

► Final Value

► Cumulative interest

► Accumulated value

► Principal value

Question No: 11 (Marks: 1)

The point where a straight line cuts the X-axis is called

► slope

► starting point

► y-intercept

► x-intercept

Question No: 12 (Marks: 1)

Which ratio is equal to 15:20?

► 21 to 28

► 5 to 10

► 18:25

► None of these

Question No: 13 (Marks: 1)

Reduction from original selling Price is called

- ▶ Loss
- ▶ List price
- ▶ Profit
- ▶ Markdown

Question No: 14 (Marks: 1)

This example returns the present value of an investment that pays Rs. 100 at the end of every year for 10 years. The money paid out will earn 5.25% annually.

- ▶ =PV (5.25%/1, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*1, 100, 1)
- ▶ =PV (5.25%/12, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*12, 100, 1)

Question No: 15 (Marks: 1)

Which is the correct syntax for the determinant of a matrix given by the following array

- ▶ =DETERM(B4:E7)
- ▶ =MDTERM(B4:E7)
- ▶ =MDETERM(B4:E7)
- ▶ =MDETERM(B4;E7)

Question No: 16 (Marks: 1)

While using Frequency function ,one always selects

▶ one cell more than data array.

▶ one cell more than bins array .

▶ at most 20 cells.

▶ random number of cells.

Question No: 17 (M a r k s: 1)

Coefficient of variation shows dispersion of the

▶ standard deviation about mean.

▶ standard deviation about mode.

▶ variance about mean.

▶ variance about mode.

Question No: 18 (M a r k s: 1)

The result of BINOMDIST is #NUM. Why?

▶ One parameter is missing.

▶ Fourth parameter is FALSE

▶ The number of successes should be negative.

▶ Probability of success on each trial should be less than 1.

Question No: 19 (M a r k s: 1)

For two tail test, when the value of Z is

▶ 1.96

▶ 1.645

▶ 2.326

▶ 2.575

Question No: 20 (Marks: 1)

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

▶ 4

▶ 1

▶ 2

▶ 3

Question No: 21 (Marks: 1)

The variable plotted on the horizontal or X-axis in a scatter diagram is called the

▶ Independent variable

▶ Dependent variable

▶ Correlation variable

▶ scatter variable

Question No: 22 (Marks: 1)

For two tail test, when the value of Z is

▶ 1.96

▶ 1.645

▶ 2.326

▶ None of these

Question No: 23 (Marks: 1)

No Linear Programming problem with an unbounded feasible region has a solution.

▶ true

▶ false

▶ may or may not

▶ none of these

Question No: 24 (Marks: 1)

The Linear Programming Model maximize or minimize the

▶ line

▶ both linear and quadratic functions

▶ None of these

▶ quadratic function

Question No: 25 (Marks: 1)

From the given scenario, Output of SUMIF(A3:A12,"Away" A3:A12) function will be

▶ 10

▶ 14

▶ 0

▶ #NAME?

Question No: 26 (Marks: 1)

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ NORMDIST

▶ NORMSINV

▶ NORMINV

Question No: 27 (Marks: 1)

There are 12 "Yes or No" questions. How many ways can these be answered?

▶ 1024

▶ 2048

► 4096

► 144

Question No: 28 (Marks: 1)

There may be more than one optimal solution of linear programming problem, however the condition is that

- the objective function must be parallel to all the constraints
- the objective function must be parallel to one of the constraints.
- the objective function must not be parallel to any of the constraints.

► None of these

Question No: 29 (Marks: 1)

For CUMULATIVE Binomial distribution, which of the following formulae is correct?

► =BINOMDIST(4, 7, 0.5, FALSE)

► =BINOMDIST(4, 7, 0.5, TRUE)

► =BINOMDIST(4, 7, 2.5, FALSE)

► =BINOMDIST(4, 7, 2.5, TRUE)

Question No: 30 (Marks: 1)

A bar graph usesto show data.

► Points

► Lines

► Circle

► Bars

Question No: 31 (Marks: 2)

How many different 5-letter arrangements are there of the letters in the word DIGIT ?

ANS:

D=1

I=2

G=1

Formula= $5 \times 4 \times 3 \times 2 \times 1 / 1 \times 2 \times 1 \times 1$

$120/2 = 60$

Question No: 32 (Marks: 2)

Explain what is meant by confidence interval?

Question No: 33 (Marks: 2)

Average sale and standard deviation for a store are 17 and 5.5 respectively .Find coefficient of variation .

ANS:

$$cv = s.d/average *100$$

$$cv=5.5/17*100$$

$$cv=32.35\%$$

Question No: 34 (Marks: 3)

Suppose a coin is flipped 3 times. What is the probability of getting two tails and one head?

ANS:

HT total probability is 50 50

now probability of getting 2 tails id 2/3

and probability of getting one head is 1/3

now total probability of getting 2 tails and 1 head is $3/6 = 1/2$

Question No: 35 (Marks: 3)

From the data below:

Week no.	Actual sales	Forecast
2	4000	4500

Given that $\alpha = 0.4$, find the forecast for the 3rd week.

Question No: 36 (Marks: 3)

A random sample of size 36 is taken from a normal population with a known variance sample is 42.6. Find the left confidence limit for the population mean.

If the mean of the

Question No: 37 (Marks: 5)

Two students were informed that they received standard scores of 0.8 and -0.4 respectively on a multiple choice examination in English. If their marks were 88 and 64 respectively. Find the mean and standard deviation of the examination marks.

Question No: 38 (Marks: 5)

Find the centered average in the data below:

Quarter	Actual	Moving Average	Centered Average
1	142		
2	54		
3	162	141	
4	206	138	139.5
1	130	137	137.5
2	50	140	138.5
3	174	138	139
4	198	137	137.5
1	126	135	136
2	42	132	133.5
3	162	129	130.5
4	186		

Question No: 39 (Marks: 5)

Find the probability that in a family of 4 children there will be at least 1 boy.

ANS:

(probability of at least boy) +(probability of girls) =1

$p(\text{boy}) = 1 - p(\text{girls})$

$p(\text{boy}) = 1 - \frac{3}{4}$

$p(\text{boy}) = \frac{1}{4}$

MIDTERM EXAMINATION
Fall 2009
MTH302- Business Mathematics & Statistics
 Solved by vuZs Team
 Mehreen Humayun
www.vuzs.net

Time: 60 min
Marks: 42

Question No: 1 (Marks: 1) - Please choose one
 45% of what is 9?

► 20

► 40

► 30

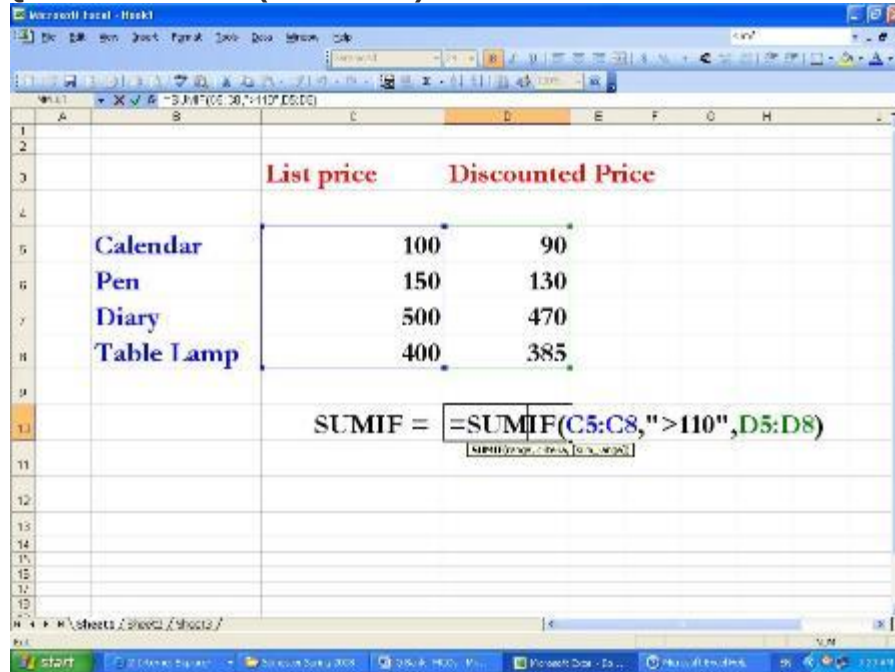
► 10

Solution:

$$45\% = 9$$

$$9 / 0.45 = 20$$

Question No: 2 (Marks: 1) - Please choose one



The screenshot shows an Excel spreadsheet with the following data:

	List price	Discounted Price
Calendar	100	90
Pen	150	130
Diary	500	470
Table Lamp	400	385

In cell D10, the formula `=SUMIF(C5:C8,">110",D5:D8)` is being entered. A tooltip for the SUMIF function is visible below the formula bar.

What will be the answer of the SUMIF formula?

► 1075

► 985

► 855

► 130

Solution:

$$=SUMIF(C5:C8,">110",D5:D8)$$

$$= 985$$

Question No: 3 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

► Sale price

► Revenue discount

► **Selling price**

► Cost price

Question No: 4 (Marks: 1) - Please choose one

In the case of partial payments, the _____ is the partial payment plus the partial cash discount

- ▶ Amount received
- ▶ Amount returned
- ▶ **Amount credited**
- ▶ Amount displaced

When you buy on credit and have cash discount terms, part of the invoice may be paid within the specified time. These part payments are called Partial Payments.

Question No: 5 (Marks: 1) - Please choose one

In a friendly cricket between India and Pakistan, Pakistan is scoring at a rate 7% more than India in 35 overs. If the India's score after 35 overs was 230 what's Pakistan score now?

- ▶ 250
- ▶ 235
- ▶ **246**
- ▶ 240

Solution:

7% of 230=16

230+16=246

Question No: 6 (Marks: 1) - Please choose one

0.20% = -----

- ▶ 0.2
- ▶ 0.02
- ▶ **0.002**
- ▶ 0.0002

Solution:

0.20/100=0.002

Question No: 7 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 2 & 6 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix}$$

, then AB is -----

- ▶ $\begin{bmatrix} 20 & 36 \\ 25 & 30 \end{bmatrix}$
- ▶ $\begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$

$$\begin{bmatrix} 20 & 36 \\ 45 & 39 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 36 \\ 35 & 30 \end{bmatrix}$$

Solution:

$$AB = \begin{bmatrix} 2 \times 1 + 6 \times 3 & 2 \times 9 + 6 \times 3 \\ 3 \times 1 + 4 \times 3 & 3 \times 9 + 4 \times 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 + 18 & 18 + 18 \\ 3 + 12 & 27 + 12 \end{bmatrix}$$

$$AB = \begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$$

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Question No: 8 (Marks: 1) - Please choose one

What will be the base of 25% rate and percentage is 65?

- ▶ 216
- ▶ 250
- ▶ **260**
- ▶ None of these

Solution:

$$\text{Rate} = 25.0 \% = 0.25$$

$$\text{Percentage} = 65$$

$$\text{Base} = 65 / 0.25 = 260$$

Question No: 9 (Marks: 1) - Please choose one

The formula for compound interest for n number of periods is

- ▶ $S = P(1 + r/100)^n$
- ▶ $S = P(1 + n/100)^r$
- ▶ **$S = P(1 + r/100)^n$**
- ▶ $S = P(1 + r/100)/n$

Compound Interest Formula

$$S = P(1 + r/100)^n$$

Question No: 10 (Marks: 1) - Please choose one

According to the local laws, the allowances are considered tax free if they are up to

- ▶ **half of the basic salary**
- ▶ one fourth of the basic salary
- ▶ one third of the basic salary
- ▶ 55% of the basic salary

Accordance to the taxation rules if allowances are 50% of basic salary, the amount is treated as tax free.

Question No: 11 (Marks: 1) - Please choose one

Amount of discount is obtained as

► **Percentage of Discount x List Price**

- Percentage of Discount / List Price
- Percentage of Discount - List Price
- None of these

Amount of discount = $d \times L$

Question No: 12 (Marks: 1) - Please choose one

In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- Provident Fund
- **Gratuity Fund**
- Stock yield
- Interest

Gratuity Fund: According to local laws, a company can establish a Gratuity Trust Fund for the benefit of the employees. By law, 1/11th of Basic Salary per month is contributed by the company to the Gratuity Fund to the account of the employee. Thus there is a saving of 1/11th of basic salary on behalf of the employee in Gratuity Fund.

Question No: 13 (Marks: 1) - Please choose one

The value of x for the equation $-(-x - 5) = x - 2$ is -----

► **no solutions**

- -5
- 3
- 1 / 2

Question No: 14 (Marks: 1) - Please choose one

What will be the simple interest earned on an amount of Rs. 16,800 in 9 months at the rate of $6\frac{1}{4}$ % p.a?

► **Rs. 787.50**

- Rs. 812.50
- Rs 860
- Rs. 887.50

Solution:

Simple interest

$I = P \times T \times R / 100$

$P=16,800$

$$T=9/12=0.75$$

$$R=6 \times 1/4=6.25$$

$$I= 16800 \times 9 \times 6.25/100$$

$$\text{Simple Interest}=787.5$$

Question No: 15 (Marks: 1) - Please choose one

This example returns the present value of an investment that pays Rs. 250 at the end of every month for 2 years. The money paid out will earn 7.5% annually.

►=PV (7.5%/12, 2*12, 250, 1)

►=PV (7.5%/12, 2*12, 250, 0)

►=PV (7.5%/12, 2, 250, 0)

►=PV (7.5%, 2*12, 250, 1)

PV(rate,nper,pmt,fv,type)

Question No: 16 (Marks: 1) - Please choose one

All formula in Excel start with

►+

►=

►-

►%

Question No: 17 (Marks: 1) - Please choose one

A percentage is a way of expressing a number as

► a fraction of 100 .

► sum of number with 100.

► multiplication of number with 100.

► average of number with 100.

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Question No: 18 (Marks: 1) - Please choose one

Given : List Price =5500Rs , Discount = 850Rs, then Net Cost Price will be

► 6350 Rs

► 5508 Rs

► 5585 Rs

► 4650 Rs

Solution:

$$\text{Net Cost Price} = \text{List price} - \text{Discount}$$

$$=55,000-850$$

$$=4,650$$

Question No: 19 (Marks: 1) - Please choose one

If C = Cost price, S = Selling Price then the margin on the cost is equal to



Markup on cost = Selling price – Cost price $\times 100\%$ / Cost Price

Question No: 20 (Marks: 1) - Please choose one

In 2 years you are to receive \$10,000. If the interest rate were to suddenly decrease, the present value of that future amount to you would

- ▶ Fall
- ▶ Rise
- ▶ remain unchanged

▶ **The correct answer cannot be determined without more information**

Question No: 21 (Marks: 1) - Please choose one

The net price of a computer table is 2500 and list price 3000. The trade discount will be

- ▶ 13%
- ▶ **17%**
- ▶ 21%
- ▶ 25%

Solution:

Trade discount = list price $\times$ discount rate

List price = 2,500

Discount rate = $2,500 - 3,000 = 500$

$500 / 3000 \times 100 = 17\%$

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Question No: 22 (Marks: 1) - Please choose one

Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10

The % Markdown is

- ▶ 10%
- ▶ **20%**
- ▶ 30%
- ▶ 40%

Solution:

Marked down = $10 / 50 \times 100 = 20\%$

Question No: 23 (Marks: 1) - Please choose one

Given for a product selling price = \$45 and unit variable cost = \$30 then Unit contribution margin will be

- ▶ \$15
- ▶ \$75

► \$1.5

► **Can not be found from given values.**

Question No: 24 (Marks: 1) - Please choose one

$3x^2 + 5x - 7$ is ----- expression.

► Monomial

► Binomial

► **Trinomial**

► Linear

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Question No: 25 (Marks: 3)

The mean of the numbers 10, 7, 9, 2 and x is 11; find the value of x.

Solution:

$$\text{Total} = 5 \times 11 = 55$$

$$\text{Mean} = 10 + 7 + 9 + 2 + x = 55$$

$$28 + x = 55$$

$$x = 55 - 28$$

$$x = 27$$

Question No: 26 (Marks: 5)

The salary of an employee is as follows:

Basic salary = 50,000 Rs.

What is the amount of allowances if House Rent = 45 %,

Conveyance allowance = 2.5 % and

Utilities allowance = 2.5 %?

Solution:

$$\text{House Rent} = 50,000 \times 45 / 100 = 22,500$$

$$\text{Conveyance Allowance} = 50,000 \times 2.5 / 100 = 1,250$$

$$\text{Utilities Allowance} = 50,000 \times 2.5 / 100 = 1,250$$

$$\text{Total allowances are } 22,500 + 1,250 + 1,250 = 25,000 \text{Rs}$$

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Question No: 27 (Marks: 10)

If the price of 16 dozen of eggs and 10 breads of large size is 332. The price of 10 dozen of eggs and 8 breads of large size is 225. Find the prices of per dozen eggs and per bread.

(Solve this question by using matrices only)

Solution:

Let we denote eggs by X and bread by Y

So equations become

Using Cramer's Rule

Therefore, price of one dozen eggs is 14.5 whereas price per bread is 10.

MIDTERM EXAMINATION

Fall 2008

MTH302- Business Mathematics & Statistics (Session - 2)

Question No: 1 (Marks: 1)

Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10

The % Markdown is

► 10%

► 20%

► 30%

► 40%

Question No: 2 (Marks: 1)

Given Net price of shirt = \$20 , Discount = 10% ,List price will

► \$22

► \$24

► \$26

► \$30

Question No: 3 (Marks: 1)

Which is the correct syntax for the determinant of a matrix given by the following array

► =DETERM(B4:E7)

► =MDTERM(B4:E7)

► =MDETERM(B4:E7)

► =MDETERM(B4;E7)

Question No: 4 (Marks: 1)

VDB returns the depreciation of an asset for

► Zero period

► One period

► Two period

► Any arbitrary period

Question No: 5 (Marks: 1)

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

► 33.33%

► 25%

► 20%

► 10%

Question No: 6 (Marks: 1)

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$8,000. The useful life of the asset is 5 years. The depreciation is being calculated for the third year, and there are 10 months in the first year.

- ▶ =DB (10000, 5, 8000, 3, 10)
- ▶ =DB (10000, 8000, 5, 10, 3)
- ▶ **=DB (10000, 8000, 5, 3, 10)**
- ▶ =DB (10000, 8000, 10, 3, 5)

Question No: 7 (Marks: 1)

Which of the following functions is not related to financial analysis?

- ▶ AMORDEGRC
- ▶ AMORLINC
- ▶ CUMIPMT
- ▶ **MDETERM**

Question No: 8 (Marks: 1)

(2) ⁻² =) **0.25**.....

- ▶ 0.35
- ▶ 0.25
- ▶ 0.45
- ▶ 0.55

Question No: 9 (Marks: 1)

Dimension of a matrix means

- ▶ Column space
- ▶ Row space
- ▶ **Order of the matrix**
- ▶ None of these

Question No: 10 (Marks: 1)

If there is a -30% change in the price of an item, what does the negative sign show?

- ▶ The price is increasing.
- ▶ **The price is decreasing.**
- ▶ The price has low rate of change.
- ▶ None of the above.

Question No: 11 (Marks: 1)

According to taxation rules; the amount of allowances is treated tax free if it is upto **50%** of the basic salary.

- ▶ Ηαλφ οφ
- ▶ Ονε τηιρδ
- ▶ 40%

► Ονε φουρτη

Question No: 12 (Marks: 1)

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

- **$I = PRT$**
- $I = P + R + T$
- $I = PT$
- $I = P(R + T)$

Question No: 13 (Marks: 1)

Every percentage can be converted into:

-⌚•Decimal
-⌚•Ratio

-⌚•Proportion
-⌚•**Both decimal & ratio**

Question No: 14 (Marks: 1)

Monthly installments of a leased car are calculated by using the techniques of:

- Simple interest method
- **Compound interest method**

- Annuity
- Hit & trial method

Question No: 15 (Marks: 1)

Suppose that you are spending 1500 dollars in an investment and Accumulated factor is 15 then Accumulated Value will be

- 22550
- **22500**
- 21000
- 22510

Question No: 16 (Marks: 1)

What is the interest on Rs. 1600 for one year at the rate %?

- 65
- **56**
- 75

Question No: 17 (Marks: 2)

A man borrows \$39000 for 1 and half year at a rate of 1.65%. What is the simple interest he has to pay?

Simple interest is given by

$$I = P \times R \times T / 100$$

P = principal = \$39000

R = interest rate = 1.65%

T = 1.5 year

$$\text{Interest, } I = (39000 \times 1.65 \times 1.5) / 100 = \text{Rs } 965.25$$

Question No: 18 (Marks: 2)

If

Cost price = 2000Rs

Selling price = 6500Rs

Then what will be %Markup on cost?

$$\text{MUC} = S - C \text{ (Rs markup on cost = selling price - cost price)}$$

$$= 6500 - 2000 = \text{Rs } 4500$$

$$\% \text{ markup on cost} = (\text{MUC} / \text{cost price}) \times 100 = (4500 / 2000) \times 100 = 225\%$$

Question No: 19 (Marks: 3)

Write the output of B9, B10, B11 and B12. If you think the formula is incorrect write "Error" in output.

ANSWER:

The output of =sum(B3,B4,B5,B6) = 100

=multiply(B5,B6) will not give result because multiply is not a excel function

B4/B6 is not a formula because in excel formula must start with "="

=B4% will return 0.2

Virtual University MCQs BANK - MCQs Collection from Online Quizzes

MATHS QUIZ 2009 Solved by (vuZs Solution Team)



Published on Sunday, 27 June 2010 12:49

Written by Bonfire

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MATHS QUIZ 2009
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Solved by (vuZs Solution Team)

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<http://groups.google.com/group/vuZs>

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Question no 1. The contribution margin per unit is an item's selling price minus the item's.....

- Fixed costs
- Variable costs
- Both variable costs & fixed costs.
- Net sales.

-

Question no 2. Contribution Margin is the amount that can be calculated by deducting Variable Cost from

- Contribution rate
- Net sale
- Net income
- **None of these.**

-

Question no.3is the value at the end of depreciation.

- **Salvage**
- Asset
- Cost
- None of these.

-

Question no.4 Difference between IRR and XIRR is that -----

- **IRR needs regular interval, but XIRR does not need regular intervals**
- XIRR does not calculate Internal Rate of Return, as IRR does.
- The schedule cash flows are needed for IRR only.
- None of these

-

Question no.5 is the number of periods over which the asset is depreciated.

- **Life**
- Asset
- Interval
- none of these

-

Question no.6 If Variable Cost is Rs.120 and contribution margin is Rs 30, then sale will be -----

- 120
- 130
- 140
- **150**

-

Question no.7 Which of the following function returns the depreciation of an asset for a specified period.

- DDB
- DB
- IRR
- MIRR

-

Question no.8 SLN is for depreciation of

- Asset
- Profit
- Loss
- Capital

-

Question no. 9 Contribution margin in Rs. Is equal to

- $=(CM/FC)$
- $=(FC/CM)*S$
- $=(S-FC)*CM$
- $= (CM-S)*FC.$